

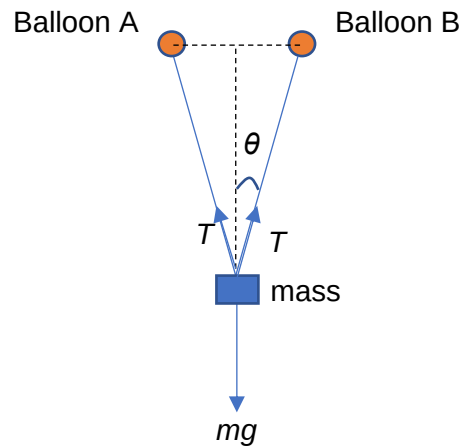
Q5(a)

Coulomb's Law states that two point charges interact with each other with a force proportional to the product of the charges and inversely proportional to the square of their separation.

[1]

[deduct mark if the word point is not included in the definition]

Q5(b)(i)



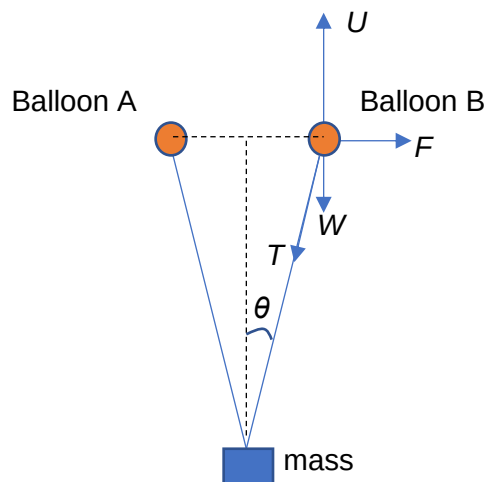
$$\sin \theta = \frac{0.30}{1.00} \Rightarrow \theta = 17.46^\circ \quad [1]$$

Consider vertical equilibrium of mass, $2T \cos \theta = mg$

$$T = \frac{mg}{2 \cos \theta} = \frac{5.0 \times 10^{-3} \times 9.81}{2 \cos 17.46^\circ} = 2.571 \times 10^{-2} \quad [1]$$

$$\approx 2.57 \times 10^{-2} \text{ N}$$

Q5(b)(ii)



U = upthrust

F = electrostatic force

W = weight of helium gas

T = tension in string

[2]

[Deduct 1 mark for missing force; 1 mark for correct labelling of forces]

Q5(b)(iii)

Resolving horizontally for equilibrium,

Coulomb's Law: $T \sin \theta = F$

$$T \sin \theta = \frac{q^2}{4\pi\epsilon_0 r^2} \text{ or } q = \sqrt{4\pi\epsilon_0 (T \sin \theta) r^2} \quad [1]$$

$$= \sqrt{(4)(3.142)(8.85 \times 10^{-12})(0.0257)(\sin 17.46^\circ)(0.60^2)} \approx \mathbf{5.56 \times 10^{-7} \text{ C}} \quad [1]$$

Q5(b)(iv)

Vertically for equilibrium, $U = W + T \cos \theta$ [1]

$$\rho_{air} Vg = \rho_{He} Vg + T \cos \theta \quad [1]$$

$$V = \frac{T \cos \theta}{(\rho_{air} - \rho_{He})g} = \frac{0.0257 \cos 17.46^\circ}{(1.29 - 0.20)(9.81)} \approx \mathbf{2.29 \times 10^{-3} \text{ m}^3} \quad [1]$$

Q6(a)